[Submission date — author to supply]

Dear Editors of Clinical Trials,

Please consider "Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence" as an original research article.

The study compares six treatment assignment designs and simple random sampling (SRS) under heterogeneous incidence and spatial spillover; two related variants are reported in the supplement. It extends our preceding BMC Medical Research Methodology publication (DOI 10.1186/s12874-026-02968-0) to a larger grid, additional assignments and a completed observed-incidence application on 58 North Carolina community-college service areas. The application evaluates estimation of a known simulated education intervention effect; it reports no implemented trial or intervention efficacy.

We assess both average mean squared error (MSE) and allocation-specific downside. Saturation performs favorably under control-only spillover, while SRS remains competitive under both-arms spillover. The manuscript explicitly reports interval undercoverage, unequal treatment budgets and finite tail uncertainty. These findings inform trial design without assuming a universally preferred allocation strategy.

The manuscript has a structured abstract and six main exhibits, with detailed reproducibility and application results in the supplement. Simulation data and code are available from the authors on request; the restricted county death-certificate data are not publicly available.

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The authors currently declare no funding and no conflicts of interest.

Sincerely,

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